Package ‘vnmap’

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Title Map the Provinces and Municipalities of Viet Nam

Version 0.1.0

Description Convenient 'ggplot2' choropleth maps of Viet Nam using either the 34 provincial-level administrative units effective from July 2025 or the preceding 63-unit geography. Includes name normalization, official administrative codes, simple-feature boundary data, and 2024 provincial population and regional-economic indicators. Boundaries are derived from the public-domain 'geoBoundaries' Viet Nam ADM1 data and the 2025 units are constructed according to Decision 19/2025/QD-TTg.

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Encoding UTF-8

LazyData true

Depends R (>= 4.1.0)

Imports ggplot2,
sf,
stringi

Suggests testthat (>= 3.0.0)

Config/testthat/edition 3

RoxygenNote 7.3.3

URL https://github.com/ura8107/vnmap

BugReports https://github.com/ura8107/vnmap/issues

Contents

vnmap-package *vnmap: Maps of Viet Nam's Provincial-Level Administrative Units*

Description

'vnmap' provides convenient 'ggplot2' choropleths, 'sf' boundary data, and administrative-code lookup tools for Viet Nam. It supports both the current 34 provincial-level units effective from 1 July 2025 and the preceding 63-unit geography used by many historical datasets.

Main functions

- [plot_vnmap()] draws outline maps and choropleths.
- [vn_map()] returns the bundled boundaries as an 'sf' object.
- [province_code()] converts names and aliases to official codes.
- [province_info()] returns the administrative lookup table.
- [vnmap_crs()] returns the package's default projected CRS.
- [province_stats_2024] contains population and GRDP-per-capita data.

Choosing a geography

Use 'geography = "provinces"' for the 34 units effective from July 2025. Use 'geography = "provinces_63"' when mapping statistics recorded under the preceding administrative structure. Always choose the geography that matches the reference period and coding scheme of the statistical data.

Data provenance

Geometry is derived from the public-domain geoBoundaries Viet Nam ADM1 boundary 'VNM-ADM1-63759600'. The current layer is constructed by dissolving historical provincial boundaries according to Decision 19/2025/QD-TTg. Source geometry is generalized and is appropriate for statistical visualization rather than surveying, navigation, or legal determinations.

Author(s)

Maintainer: Yuki Matsuura `<matsuura.yuki.200095@gmail.com>`

References

geoBoundaries: https://www.geoboundaries.org/

See Also

Useful links:

- https://github.com/ura8107/vnmap
- Report bugs at https://github.com/ura8107/vnmap/issues

plot_vnmap *Plot a map of Viet Nam*

Description

Creates a ‘ggplot2‘ outline map or choropleth from the bundled provincial boundaries. Statistical values are joined using names, aliases, official codes, or historical codes appropriate to the selected geography.

Usage

```
plot_vnmap(
  geography = c("provinces", "provinces_63"),
  data = NULL,
  values = NULL,
  id = "code",
  include = NULL,
  labels = FALSE,
  ...
)
```

Arguments

geography	Either ‘"provinces"‘ for the current 34-unit geography or ‘"provinces_63"‘ for the historical 63-unit geography.
data	Optional data frame containing one row per administrative unit.
values	Character scalar naming the column in ‘data‘ mapped to the polygon fill aesthetic. Required when ‘data‘ is supplied.
id	Character scalar naming the column in ‘data‘ containing province or municipality names, aliases, two-digit codes, or ISO codes.
include	Optional character vector of unit names or codes to display.
labels	Logical; if ‘TRUE‘, add Vietnamese names at points guaranteed to lie inside each geometry.
...	Passed to [ggplot2::geom_sf()].

Details

When ‘data‘ is supplied, units without a matching row receive ‘NA‘ as the fill value and are handled by the chosen fill scale. Duplicate identifiers in ‘data‘ are rejected because a polygon can have only one fill value. Unit identifiers are normalized by [province_code()].

Arguments such as ‘color‘, ‘linewidth‘, and ‘alpha‘ are passed through to [ggplot2::geom_sf()]. Labels use ‘name_vi‘ and may overlap on dense maps; ‘check_overlap = TRUE‘ suppresses some overlapping labels.

Value

A ‘ggplot‘ object. Add scales, titles, annotations, and themes using ordinary ‘ggplot2‘ layers.

See Also

[vn_map()], [province_code()], [ggplot2::geom_sf()]

Examples

```
plot_vnmap(color = "white", linewidth = 0.2)

values <- transform(province_info(), value = seq_len(34))
plot_vnmap(
  data = values,
  values = "value",
  id = "code",
  color = "white",
  linewidth = 0.2
) + ggplot2::scale_fill_viridis_c(name = "Example")

plot_vnmap(
  geography = "provinces_63",
  include = c("Ha Noi", "Hai Phong", "Quang Ninh"),
  labels = TRUE
)
```

`province_code` *Look up Vietnamese provincial administrative codes*

Description

Names may be written with or without Vietnamese diacritics. Current names, English names, two-digit codes, and ISO 3166-2 codes are accepted.

Usage

```
province_code(x, geography = c("provinces", "provinces_63"))
```

Arguments

`x`	Character vector of names or codes.
`geography`	Either ‘"provinces"‘ for the current 34-unit geography or ‘"provinces_63"‘ for the historical 63-unit geography.

Details

Input is matched case-insensitively after punctuation, whitespace, administrative prefixes, and Vietnamese diacritics are normalized. Common aliases such as ‘"Hanoi"‘, ‘"Danang"‘, ‘"HCMC"‘, and ‘"Saigon"‘ are supported. An error lists any values that cannot be matched.

Codes are geography-specific. For example, a former province that was merged in 2025 can be found only with ‘geography = "provinces_63"‘.

Value

A character vector of two-digit administrative codes.

See Also

[province_info()], [vn_map()]

Examples

```
province_code(c("Da Nang", "Danang", "48"))
province_code(c("HCMC", "Saigon"))
province_code("Bac Giang", geography = "provinces_63")
```

`province_info` *Retrieve metadata for Vietnamese provincial units*

Description

Returns the lookup table used by ‘vnmap‘ to join statistical data to map geometry. Supplying ‘x‘ filters and orders the result to match the input.

Usage

```
province_info(x = NULL, geography = c("provinces", "provinces_63"))
```

Arguments

`x`	Optional names or codes. When omitted, returns every unit.
`geography`	Either ‘"provinces"‘ for the current 34-unit geography or ‘"provinces_63"‘ for the historical 63-unit geography.

Details

With no ‘x‘, rows are returned in ascending official code order. With ‘x‘, names and aliases are normalized by [province_code()] and the returned rows follow the order of ‘x‘.

Value

A data frame containing ‘code‘, ‘iso‘, ‘name_vi‘, ‘name_en‘, and ‘type‘. The ‘type‘ column distinguishes provinces from centrally governed municipalities.

See Also

[province_code()], [vn_map()]

Examples

```
head(province_info())
province_info(c("01", "HCMC"))
province_info("Bac Giang", geography = "provinces_63")
```

`province_stats_2024` *Provincial population and economic indicators, 2024*

Description

Official 2024 area, average population, population density, and gross regional domestic product (GRDP) per capita for the current 34 provincial units of Viet Nam. Values for units created in 2025 are aggregated from the 63-unit geography in which the statistics were published.

Usage

```
province_stats_2024
```

Format

A data frame with 34 rows and 12 variables:

code Current two-digit administrative code.

name_vi, name_en Vietnamese and English unit names.

type Province or centrally governed municipality.

year Reference year, 2024.

area_km2 Land area in square kilometres.

population Average population, in persons.

population_thousands Average population, in thousands of persons.

population_density Population divided by area, persons per km2.

grdp_per_capita_million_vnd GRDP per capita, million current VND.

grdp_preliminary Whether the GRDP value is preliminary.

member_codes Codes of the pre-2025 units used in aggregation.

Details

Population and area are summed across member units. Population density is recalculated from those totals. GRDP per capita is population- weighted, which is equivalent to summing estimated provincial GRDP and dividing by the combined population. Because published inputs are rounded, aggregated figures should be treated as estimates.

Source

General Statistics Office of Viet Nam, PX-Web tables "Area, population and population density by province" and "Gross regional domestic product per capita by province", accessed 13 July 2026. https://pxweb.nso.gov.vn/

Examples

```
data(province_stats_2024)
head(province_stats_2024)
plot_vnmap(province_stats_2024, "grdp_per_capita_million_vnd", id = "code")
```

`province_stats_2024_63` *Provincial population and economic indicators under the 63-unit geography*

Description

The official 2024 source values before aggregation to Viet Nam's current 34-unit provincial geography.

Usage

```
province_stats_2024_63
```

Format

A data frame with 63 rows and 12 variables. Variables correspond to those in [province_stats_2024], except that ‘iso‘ replaces ‘member_codes‘.

Source

General Statistics Office of Viet Nam, PX-Web tables, accessed 13 July 2026. https://pxweb.nso.gov.vn/

Examples

```
data(province_stats_2024_63)
plot_vnmap(
  geography = "provinces_63",
  data = province_stats_2024_63,
  values = "population_density",
  id = "code"
)
```

`vn_map` *Obtain Viet Nam map data*

Description

Returns a simple-features boundary layer for the provincial-level administrative units of Viet Nam. The current layer contains the 34 units effective from 1 July 2025. A 63-unit layer is retained for data collected under the immediately preceding geography.

Usage

```
vn_map(
  geography = c("provinces", "provinces_63"),
  include = NULL,
  crs = vnmap_crs()
)
```

Arguments

geography	Either ‘"provinces"‘ (the 34 units effective from July 2025) or ‘"provinces_63"‘ (the pre-July 2025 units).
include	Optional character vector of names, aliases, official two-digit codes, or ISO 3166-2 codes to retain. Vietnamese names may be supplied with or without diacritics.
crs	Coordinate reference system understood by [sf::st_transform()].

Details

The bundled geometry is projected to EPSG:3405 by default. Pass a different EPSG code or an ‘sf‘ CRS object to ‘crs‘ when another projection is required. Filtering happens before transformation.

The 34-unit layer was constructed by dissolving historical provincial boundaries according to Decision 19/2025/QD-TTg. The source geometry is generalized and intended for statistical graphics, not surveying, navigation, or legal boundary determinations.

Value

An ‘sf‘ object with one row per unit and the following columns: ‘code‘, ‘iso‘, ‘name_vi‘, ‘name_en‘, ‘type‘, and ‘geometry‘. Current units created by the 2025 reform have ‘NA‘ in ‘iso‘ where no code is bundled.

See Also

[plot_vnmap()], [province_code()], [province_info()]

Examples

```
map <- vn_map()
nrow(map)
names(map)

north <- vn_map(include = c("Ha Noi", "Bac Ninh"))
nrow(north)

historical <- vn_map("provinces_63", crs = 4326)
sf::st_crs(historical)$input
```

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vnmap_crs *Coordinate reference system used by vnmap*

Description

EPSG:3405 is the VN-2000 / UTM zone 48N projected CRS.

Usage

```
vnmap_crs()
```

Details

All bundled geometries are returned in this projected CRS unless a different value is supplied to the ‘crs‘ argument of [vn_map()]. Using a projected CRS keeps distances and polygon rendering stable for national statistical maps.

Value

An ‘sf‘ CRS object.

See Also

[vn_map()], [sf::st_crs()]

Examples

```
vnmap_crs()
```

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